

SIMULATED INCIDENT RESPONSE & DFIR CASE STUDY

Development Environment Data Breach

Classification	Portfolio / Training Simulation
Role	Incident Response Analyst / Incident Commander
Environment	Fictional enterprise development network
Incident Type	Credential Theft, Session Hijacking, Lateral Movement, Persistence, Data Exfiltration
Status	Known intrusion path contained; external attribution handed off

Disclaimer: This is a fictional cybersecurity training simulation created for educational and portfolio purposes. It does not represent access to, participation in, or investigation of any real Rockstar Games systems or incident.

1. Executive Summary

This simulated incident-response investigation reconstructed a multi-stage compromise that resulted in the exfiltration of approximately 14.8 GB of confidential development data from a fictional enterprise environment.

The intrusion began with a phishing attack against an employee who accessed Microsoft 365 from a personally owned device. A malicious information-stealing executable captured browser authentication/session artifacts after the employee legitimately completed MFA. The attacker later reused the compromised authentication context to access the corporate VPN.

A temporary contractor exception had left managed-device enforcement in audit-only mode rather than enforced mode. This did not cause the original credential compromise, but it removed an important defense-in-depth control and allowed an unmanaged attacker-controlled device to establish access.

Using the compromised identity, the attacker reached DEPLOY-02, obtained a development service account from credential resources, moved laterally through development servers, established persistence, compromised additional users, reached BUILD-SRV-04, staged sensitive data, compressed it into a 14.8 GB archive, and exfiltrated the archive over HTTPS to attacker-controlled infrastructure.

The investigation identified and contained the known initial-access path, compromised sessions, attacker-created accounts, malicious persistence mechanisms, and security configuration changes. External attribution was preserved and handed off to Legal and law enforcement.

2. Investigation Objectives

- Determine the initial access vector
- Identify compromised identities and endpoints
- Reconstruct lateral movement and privilege expansion
- Identify persistence mechanisms
- Determine the source and scope of data access
- Identify staging and exfiltration mechanisms
- Contain known attacker access
- Preserve evidence for Legal and law enforcement
- Identify root causes and hardening priorities

3. Reconstructed Attack Path



Figure 1. Reconstructed intrusion path for the simulated incident.

4. Key Incident Timeline

Stage	Representative Time	Identity/System	Finding
Initial access	21 days before intrusion	Thompson personal laptop	Phishing payload captures browser/session material
VPN access	Incident day	r.thompson	Compromised session

			accepted from unmanaged device due to audit-only control
Internal access	Incident day	DEPLOY-02	Attacker reaches deployment environment and credential resources
Lateral movement	03:02-03:41	svc_devops	Successful DEV server access; FIN-SRV-02, DC-01, BUILD-SRV-04 denied
Persistence	During DEV activity	dev_support / backup_operator	Unauthorized accounts and scheduled task established
Secondary compromise	04:12+	Rodriguez workstation	Remote service and credential-memory access
Trusted phishing	08:54+	Rodriguez -> Chen	Compromised mailbox delivers malicious ZIP to Chen
Build server	09:07+	Chen -> BUILD-SRV-04	Successful access, discovery, staging and compression
Exfiltration	10:04-10:51	BUILD-SRV-04	~14.8 GB transferred over HTTPS to attacker VPS

5. Detailed Findings

5.1 Initial Access and Session Theft

Thompson received a phishing message on a personally owned laptop used for Microsoft 365 access. The message impersonated IT support and led to `compatibility_update.zip` containing `BrowserCompatibilityUpdate.exe`.

Reverse engineering showed browser/session collection behavior rather than a full remote-access trojan. The payload accessed Chromium profile data, Microsoft authentication-related storage, and session artifacts, then transmitted a small encrypted package externally.

Thompson later completed a legitimate Microsoft 365 login and MFA. The evidence supported theft of persistent session/refresh material rather than theft of Thompson's plaintext password or direct MFA bypass.

5.2 VPN Control Failure

The corporate VPN normally required a managed company device. A temporary contractor exception changed the control to audit-only for 14 days but remained active for approximately six months.

The attacker presented a valid compromised authentication context from an unmanaged device. The compliance failure was logged but not blocked, allowing network access. The forgotten exception was a major control failure, but it is treated as an impact-amplifying condition rather than the root cause of the original credential compromise.

5.3 DEPLOY-02 and Service Account Exposure

Using Thompson's access, the attacker reached DEPLOY-02 and credential resources associated with development infrastructure. The attacker obtained access to svc_devops, which had broader reach across multiple development servers.

5.4 DEV Infrastructure Activity

DEV-SRV-10 showed authentication and reconnaissance with no persistence. DEV-SRV-11 contained a suspicious short-lived executable used for credential/session discovery and outbound encrypted communication. DEV-SRV-12 showed account/group enumeration and failed account-creation attempts. DEV-SRV-13 was used for network reconnaissance and pivoting. DEV-SRV-14 had firewall scope widened, logging reduced, and Defender network protection changed from Block to Audit.

svc_devops successfully accessed DEV-SRV-10, 11, 12, 13 and 14, but was denied access to FIN-SRV-02, DC-01 and BUILD-SRV-04.

5.5 Persistence

The attacker created dev_support and then backup_operator. The chain stopped at backup_operator; review of the relevant timeframe found no additional successfully created attacker accounts.

backup_operator established host persistence on DEV-SRV-12 through the DevelopmentUpdateCheck scheduled task, which launched devsync.exe as SYSTEM. The accounts, scheduled task and executable were later disabled or removed and relevant sessions/tickets revoked.

5.6 Rodriguez Compromise

From DEV-SRV-14, the attacker used svc_devops over SMB/ADMIN\$ to place healthcheck.exe on Rodriguez's workstation and create the SystemHealthMonitor service. The payload accessed credential-related memory, and Rodriguez's authentication context was subsequently observed being used from DEV-SRV-14 while she remained on her own workstation.

5.7 Chen Compromise

The attacker used Rodriguez's legitimate corporate mailbox to send Chen an Updated Project Schedule email containing Updated_Project_Schedule.zip. Chen manually extracted and launched ProjectSchedule.exe.

ProjectSchedule.exe performed user/system enumeration, credential/session discovery, internal network discovery and command-and-control. It established persistence using UpdateScheduleCheck and schedulehost.exe. The malware identified that Chen had legitimate access to build-related infrastructure that svc_devops could not reach.

5.8 BUILD-SRV-04 Collection and Exfiltration

The attacker used Chen's compromised context to access BUILD-SRV-04. Directory and file-access logs showed discovery of Rockstar project, release-candidate and internal design resources. A small number of representative image, PDF and video files were opened to validate the value of the material before bulk collection.

Selected data was staged under C:\ProgramData\BuildCache and compressed with a portable 7-Zip executable into cache.7z, approximately 14.8 GB. PowerShell and network telemetry showed the archive uploaded over HTTPS directly from BUILD-SRV-04 to attacker-controlled external infrastructure. Cleanup commands removed the archive and staging artifacts, but endpoint, NTFS, PowerShell and network evidence remained sufficient to reconstruct the operation.

6. Containment and Eradication

- Revoked Thompson's active sessions and refresh/session material; temporarily disabled and reset affected credentials.
- Restored VPN managed-device compliance from Audit Only to Enforced.
- Removed dev_support and backup_operator, revoked their sessions/tickets, and removed elevated memberships.
- Disabled DevelopmentUpdateCheck and quarantined devsync.exe; validated DEV-SRV-12 for additional persistence.
- Isolated/remediated Rodriguez and Chen workstations and invalidated affected credentials/sessions.
- Identified and reversed unauthorized security-control changes on DEV-SRV-14.
- Preserved endpoint, identity, email, VPN, firewall, server and exfiltration evidence for investigation.
- Communicated to leadership that the known intrusion path was contained without overstating a complete all-clear.

7. Root Causes and Contributing Factors

1. Successful phishing and execution of an untrusted executable on a personal device.
2. Corporate authentication permitted from an unmanaged BYOD endpoint.
3. Persistent authentication/session material remained reusable for an extended period.
4. Temporary VPN exception lacked reliable expiration and automatic reversion.
5. svc_devops possessed excessive reach across development infrastructure.
6. Credential resources accessible from DEPLOY-02 enabled privilege expansion.
7. Internal administrative protocols enabled lateral movement between development systems.
8. Security alerts for credential access and suspicious executable activity were not escalated quickly enough.
9. Trusted internal email made secondary phishing highly effective.
10. Large outbound archive transfer was not blocked before exfiltration completed.

8. Recommended Hardening

- Enforce managed-device requirements for VPN and sensitive cloud applications.
- Implement automatic expiration for temporary security exceptions with owner, approval and rollback date.
- Restrict BYOD access and require stronger device posture for privileged or sensitive workloads.
- Reduce session lifetime and require risk-based reauthentication for sensitive systems.
- Improve phishing, attachment and executable controls; block execution from user-writable locations where feasible.
- Reduce service-account privilege and segment service-account access by application and server role.
- Harden credential vault access and rotate service credentials following exposure.
- Restrict SMB admin shares, remote service creation and east-west administrative traffic.

- Improve alerting for new privileged accounts, scheduled tasks, service creation, LSASS access and security-control downgrades.
- Implement DLP/egress monitoring for large archive creation and sustained outbound uploads.
- Conduct targeted security-awareness training after containment and root-cause validation.

9. MITRE ATT&CK Mapping

Technique	Name	Scenario Evidence
T1566	Phishing	Initial malicious message to Thompson and later trusted-account phishing to Chen
T1204	User Execution	Thompson and Chen manually executed malicious payloads
T1550	Use Alternate Authentication Material	Attacker reused compromised authenticated session material
T1078	Valid Accounts	Thompson, svc_devops, Rodriguez and Chen contexts abused
T1021	Remote Services	SMB/admin access and remote service creation used for lateral movement
T1003	OS Credential Dumping / Credential Access	Credential-related memory access on compromised endpoints
T1087	Account Discovery	Account and group enumeration on DEV infrastructure
T1018	Remote System Discovery	Internal host discovery and build-system identification
T1136	Create Account	dev_support and backup_operator persistence accounts
T1053	Scheduled Task/Job	DevelopmentUpdateCheck and UpdateScheduleCheck persistence
T1562	Impair Defenses	Firewall/logging/Defender policy weakened on DEV-SRV-14
T1074	Data Staged	Sensitive files staged under BuildCache
T1560	Archive Collected Data	7-Zip used to create cache.7z
T1041	Exfiltration Over C2/Web Service	Archive uploaded externally over HTTPS

10. Skills Demonstrated

- Incident response lifecycle and containment prioritization
- Identity, MFA and session-token analysis
- VPN and conditional-access control analysis
- EDR and Windows event investigation
- Lateral movement and service-account abuse analysis
- Persistence and Active Directory account auditing
- Email/phishing and trusted-account compromise analysis
- Malware behavior and reverse-engineering interpretation
- Network and exfiltration reconstruction

- Evidence preservation, privacy and Legal escalation
- Executive communication and root-cause analysis

11. Portfolio Positioning

Suggested résumé project title: Incident Response & DFIR Simulation - Enterprise Development Environment

- Led a simulated end-to-end incident response investigation involving phishing, session theft, VPN control failure, service-account abuse, lateral movement, persistence and 14.8 GB data exfiltration.
- Correlated EDR, Windows, IAM, VPN, firewall, email and network telemetry to reconstruct the attack path and identify compromised identities and systems.
- Investigated malicious executables, credential access, attacker-created accounts, scheduled-task persistence, security-control changes, data staging and HTTPS exfiltration.
- Produced executive and technical incident summaries while incorporating containment, evidence preservation, privacy and Legal escalation considerations.

12. Disclaimer

This case study is entirely fictional and was created as a cybersecurity training simulation for portfolio and résumé purposes. It does not represent participation in or access to any actual Rockstar Games investigation, systems, data, breach, or confidential information.